

Stat 286: Causal Inference with Applications, Fall 2025

Section 3: Average Treatment Effects

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1 Overview

We distinguish two complementary perspectives for average treatment effects (ATEs): the finite-population (design-based) view and the super-population (model/sampling-based) view.

Finite-population (design-based) view. We fix the n units in our study. By unit-level determinism (SUTVA/consistency assumed), each unit i has fixed potential outcomes $\{Y_i(0), Y_i(1)\}$; the only randomness comes from the assignment mechanism. The causal estimand of interest is the *sample* ATE (SATE),

$$\tau_{\text{SATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)).$$

Super-population (model-based) view. We imagine the units are an i.i.d. sample from a large (possibly infinite) population where $(Y(1), Y(0)) \sim \mathbb{P}^*$ are random. Randomness comes from the treatment assignment and the sampling. The causal estimand of interest is the *population* ATE (PATE)

$$\tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)].$$

2 Comparison

3 Results

LEMMA 1 (UNBIASEDNESS OF THE DIFFERENCE-IN-MEANS ESTIMATOR) Let $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$, where $\bar{Y}_t = \frac{1}{n_t} \sum_{i:T_i=t} Y_i$ and $n_t = \sum_{i=1}^n \mathbf{1}\{T_i = t\}$.

- (i) (**Finite population / SATE**) Under a completely randomized design (CDR) with fixed n_1 and n_0 , conditioning on the fixed potential outcomes $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$,

$$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{\text{SATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)).$$

- (ii) (**Super-population / PATE**) If, in addition, the n units are drawn as an i.i.d. simple random sample from a population where $(Y(1), Y(0))$ are random, then

$$\mathbb{E}[\hat{\tau}] = \tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)].$$

REMARK 1 We also have that $\mathbb{E}[\hat{\sigma}_t^2] = \sigma_t^2$ by the usual unbiasedness of the sample variance.

Proof: (i) *Finite population.* Using linearity of expectation and the fact that the potential outcomes are fixed given $\mathcal{O}_n$,

$$\mathbb{E}[\bar{Y}_1 \mid \mathcal{O}_n] = \mathbb{E}\left[\frac{1}{n_1} \sum_{i=1}^n T_i Y_i(1) \mid \mathcal{O}_n\right] = \frac{1}{n_1} \sum_{i=1}^n Y_i(1) \mathbb{E}[T_i \mid \mathcal{O}_n] = \frac{1}{n_1} \sum_{i=1}^n Y_i(1) \frac{n_1}{n} = \frac{1}{n} \sum_{i=1}^n Y_i(1).$$

The equality follows because, by symmetry of the CRD with fixed n_1 , each i is selected into treatment with probability n_1/n ,

$$\mathbb{E}[T_i \mid \mathcal{O}_n] = \mathbb{E}[T_i] = \Pr(T_1 = 1) = \frac{n_1}{n}$$

Analogously,

$$\mathbb{E}[\bar{Y}_0 \mid \mathcal{O}_n] = \frac{1}{n} \sum_{i=1}^n Y_i(0).$$

Subtracting yields

$$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0)) = \tau_{\text{SATE}}.$$

- (ii) *Super-population.* Using the law of iterated expectation and (i),

$$\mathbb{E}[\hat{\tau}] = \mathbb{E}[\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n]] = \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))\right] = \mathbb{E}[Y(1) - Y(0)] = \tau_{\text{PATE}},$$

where the last equality uses i.i.d. sampling so that the expectation of the sample mean equals the population mean. $\square$

LEMMA 2 (VARIANCE OF THE DIFFERENCE-IN-MEANS ESTIMATOR) Let $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$, where $\bar{Y}_t = \frac{1}{n_t} \sum_{i:T_i=t} Y_i$ and $n_t = \sum_{i=1}^n \mathbf{1}\{T_i = t\}$, under a completely randomized design (CRD) with fixed n_1 and n_0 .

- (i) (**Finite population / SATE**) Conditioning on $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$,

$$\text{Var}(\hat{\tau} \mid \mathcal{O}_n) = \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n} \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0},$$

with $S_*^2 = S_1^2 + S_0^2 - 2S_{01} \geq 0$ and $\mathbb{E}[S_*^2] = \text{Var}(Y_i(1) - Y_i(0))$.

- (ii) (**Super-population / PATE**) If, in addition, the n units are drawn i.i.d. from a population where $(Y(1), Y(0))$ are random,

$$\text{Var}(\hat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}.$$

REMARK 2 *The above lemma illustrates why we can use the estimator for the super-population variance for the finite-population variance. In expectation this estimator yields.*

$$\mathbb{E}[\widehat{\text{Var}}(\hat{\tau}) \mid \mathcal{O}_n] = \mathbb{E}\left[\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}\right] = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} \geq \text{Var}(\hat{\tau} \mid \mathcal{O}_n)$$

Proof: (i) *Finite population.* The first equality follows from the review questions. We show the second equality here. By definition $2S_{01} = S_1^2 + S_0^2 - S_*^2$. Inserting into the variance

$$\begin{aligned} \text{Var}(\hat{\tau} \mid \mathcal{O}_n) &= \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right) \\ &= \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + S_1^2 + S_0^2 - S_*^2 \right) \\ &= \frac{1}{n} \left(\frac{n_0 + n_1}{n_1} S_1^2 + \frac{n_1 + n_0}{n_0} S_0^2 - S_*^2 \right) \\ &= \frac{1}{n} \left(\frac{n}{n_1} S_1^2 + \frac{n}{n_0} S_0^2 - S_*^2 \right) \\ &= \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n}. \end{aligned}$$

This proves the second equality. It remains to show that $S_*^2 \geq 0$. By Cauchy-Schwarz we have

$$S_{01} \leq S_1 S_0.$$

Noting that $S_t \geq 0$ we have

$$S_*^2 = S_1^2 + S_0^2 - 2S_{01} \geq S_1^2 + S_0^2 - 2S_1 S_0 = (S_1 - S_0)^2 \geq 0,$$

which proves the last claim.

(ii) *Super-population.* By the law of iterated variance and our i.i.d. assumption:

$$\begin{aligned} \text{Var}(\hat{\tau}) &= \mathbb{E}[\text{Var}(\hat{\tau} \mid \mathcal{O}_n)] + \text{Var}(\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n]) \\ &= \mathbb{E}\left[\frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_*^2}{n}\right] + \text{Var}\left(\frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))\right) \\ &= \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0} - \frac{1}{n} E[S_*^2] + \frac{1}{n} \text{Var}(Y_i(1) - Y_i(0)) \end{aligned}$$

since $\mathbb{E}[S_t^2] = \sigma_t^2$ by the unbiasedness of the sample variance. It remains to show that $E[S_*^2] = \text{Var}(Y(1) - Y(0))$. Let $\mu_1 = \mathbb{E}[Y(1)]$, $\mu_0 = \mathbb{E}[Y(0)]$. Expand

$$S_{01} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \overline{Y(1)})(Y_i(0) - \overline{Y(0)}) = \frac{1}{n-1} \left[\sum_{i=1}^n (Y_i(1) - \mu_1)(Y_i(0) - \mu_0) - n(\overline{Y(1)} - \mu_1)(\overline{Y(0)} - \mu_0) \right]$$

Taking expectations,

$$\begin{aligned} \mathbb{E}\left[\sum_{i=1}^n (Y_i(1) - \mu_1)(Y_i(0) - \mu_0)\right] &= n \text{Cov}(Y_i(1), Y_i(0)) \\ \mathbb{E}\left[n(\overline{Y(1)} - \mu_1)(\overline{Y(0)} - \mu_0)\right] &= n \text{Cov}(\overline{Y(1)}, \overline{Y(0)}) \\ &= n \frac{1}{n^2} \sum_{i,j=1}^n \text{Cov}(Y_i(1), Y_j(0)) \\ &= \frac{1}{n} \sum_{i=1}^n \text{Cov}(Y_i(1), Y_i(0)) \\ &= \text{Cov}(Y_i(1), Y_i(0)) \end{aligned}$$

Therefore

$$\mathbb{E}[S_{01}] = \frac{1}{n-1}(n-1)\text{Cov}(Y_i(1), Y_i(0)) = \text{Cov}(Y_i(1), Y_i(0))$$

By linearity of expectation,

$$\mathbb{E}[S_*^2] = \mathbb{E}[S_1^2] + \mathbb{E}[S_0^2] - 2\mathbb{E}[S_{01}] = \sigma_1^2 + \sigma_0^2 - 2\text{Cov}(Y_i(1), Y_i(0)).$$

Finally, for any two random variables A, B , $\text{Var}(A - B) = \text{Var}(A) + \text{Var}(B) - 2\text{Cov}(A, B)$. With $A = Y(1)$ and $B = Y(0)$,

$$\text{Var}(Y(1) - Y(0)) = \sigma_1^2 + \sigma_0^2 - 2\text{Cov}(Y_i(1), Y_i(0)).$$

Thus $\mathbb{E}[S_*^2] = \text{Var}(Y(1) - Y(0))$, as claimed.

□

	Finite-Population	Super-Population
<i>Population</i>	Set of units in the study is fixed and finite	Units are drawn i.i.d. from a large or infinite “super-population.”
<i>Potential Outcomes</i>	Fixed set $\mathcal{O}_n = \{Y_i(0), Y_i(1)\}_{i=1}^n$	$(Y_i(0), Y_i(1))$ i.i.d. $\sim \mathbb{P}^*$
<i>Inference</i>	Treatment effects for this specific group of units	Treatment effects for the underlying population distribution
<i>Source of Randomness</i>	Treatment assignment	Treatment assignment Sampling of units from the population
<i>Random Variables</i>	Treatment T_i	Treatment and Potential Outcomes $T_i, Y_i(0), Y_i(1)$
<i>Example</i>	What is the ATE on these n units in our experiment?	What is the ATE in the population from which these n units were sampled?
<i>Validity</i>	Internal	External
<i>Estimand</i>	$\tau_{\text{SATE}} = \frac{1}{n} \sum_{i=1}^n (Y_i(1) - Y_i(0))$	$\tau_{\text{PATE}} = \mathbb{E}[Y(1) - Y(0)]$
<i>Estimator</i>	Difference in means $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$	
<i>Unbiasedness</i>	$\mathbb{E}[\hat{\tau} \mid \mathcal{O}_n] = \tau_{\text{SATE}}$	$\mathbb{E}[\hat{\tau}] = \tau_{\text{PATE}}$
<i>Variance of Estimator</i>	$\text{Var}(\hat{\tau} \mid \mathcal{O}_n) = \frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right)$	$\text{Var}(\hat{\tau}) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}$
<i>Estimator of Variance</i>	$\widehat{\text{Var}}(\hat{\tau}) = \frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}$ Finite-Population: Conservative estimator $\mathbb{E}[\widehat{\text{Var}}(\hat{\tau}) \mid \mathcal{O}_n] \geq \text{Var}(\hat{\tau} \mid \mathcal{O}_n)$, equal to design variance iff $Y_i(1) - Y_i(0) = c$ for all i , Super-Population: Unbiased estimator $\mathbb{E}[\widehat{\text{Var}}(\hat{\tau})] = \text{Var}(\hat{\tau})$	
<i>Asymptotics</i> $\frac{n_1}{n} \rightarrow p \in (0, 1)$	Finite-population CLT: $\frac{\hat{\tau} - \tau_{\text{SATE}}}{\sqrt{\frac{1}{n} \left(\frac{n_0}{n_1} S_1^2 + \frac{n_1}{n_0} S_0^2 + 2S_{01} \right)}} \xrightarrow{d} \mathcal{N}(0, 1)$	Super-population LLN and CLT: Consistency (LLN): $\hat{\tau} \xrightarrow{p} \tau_{\text{PATE}}$ CLT: $\frac{\hat{\tau} - \tau_{\text{PATE}}}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_0^2}{n_0}}} \xrightarrow{d} \mathcal{N}(0, 1)$
<i>CI in practice</i>	s.e. = $\sqrt{\frac{\hat{\sigma}_1^2}{n_1} + \frac{\hat{\sigma}_0^2}{n_0}}$ $(1 - \alpha) \times 100\%$ CI: $[\hat{\tau} \pm z_{1-\alpha/2} \cdot \text{s.e.}]$ Conservative in finite-population; Unbiased in super-population	

Assumptions

The Table is valid under SUTVA and a completely randomized design (CRD).

Notation

$$n_1 = \sum_{i=1}^n T_i, \quad n_0 = n - n_1$$

$$\overline{Y(t)} = \frac{1}{n} \sum_{i=1}^n Y_i(t), \quad \bar{Y}_t = \frac{1}{n_t} \sum_{i:T_i=t} Y_i,$$

$$S_t^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i(t) - \overline{Y(t)})^2, \quad S_{01} = \frac{1}{n-1} \sum_{i=1}^n (Y_i(1) - \overline{Y(1)})(Y_i(0) - \overline{Y(0)}),$$

$$\sigma_t^2 = \text{Var}(Y_i(t)) = \mathbb{E}[S_t^2], \quad \hat{\sigma}_t^2 = \frac{1}{n_t - 1} \sum_{i:T_i=t} (Y_i - \bar{Y}_t)^2.$$

Finite-Population Bounds for the Variance using Cauchy Schwarz

Since S_{01} is unobserved, we can obtain these (tight without further assumptions) bounds on the variance for the τ_{SATE}

$$\frac{n_0 n_1}{n} \left(\frac{S_1}{n_1} - \frac{S_0}{n_0} \right)^2 \leq \text{Var}(\hat{\tau} \mid \mathcal{O}_n) \leq \frac{n_0 n_1}{n} \left(\frac{S_1}{n_1} + \frac{S_0}{n_0} \right)^2$$